

# Reading off a surface-code distance

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The code distance  $d$  of a planar surface code is just the length of the shortest logical operator — the shortest chain of data qubits connecting the two like-type boundaries. Eyeballing it on the lattice is usually faster than any formula.

A distance- $d$  code corrects up to  $\lfloor (d-1)/2 \rfloor$  arbitrary errors, so distance grows the protection roughly exponentially *below threshold*: doubling  $d$  squares the suppression of logical error. The catch is cost — a planar patch needs on the order of  $d^2$  physical qubits, so every step up in distance is quadratically expensive.

Practical takeaway: pick the smallest  $d$  whose projected logical error rate clears your algorithm's total operation budget, not the largest one you can fit.